Bone Scintigraphy
Lec“1”

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Learning Objectives

By the end of this lecture, the technologist should be able to:

- ❑ Explain the principle of bone scintigraphy.
- ❑ Describe radiopharmaceuticals used for bone imaging.
- ❑ Prepare and administer the radiopharmaceutical safely.
- ❑ Perform the three phases of bone scintigraphy.
- ❑ Understand patient preparation and imaging protocols.
- ❑ Recognize normal skeletal distribution.
- ❑ Identify common abnormal patterns.
- ❑ Understand SPECT/SPECT-CT applications.





Bone scintigraphy

Bone scintigraphy is a nuclear medicine imaging technique used to evaluate osteoblastic activity and regional bone turnover.



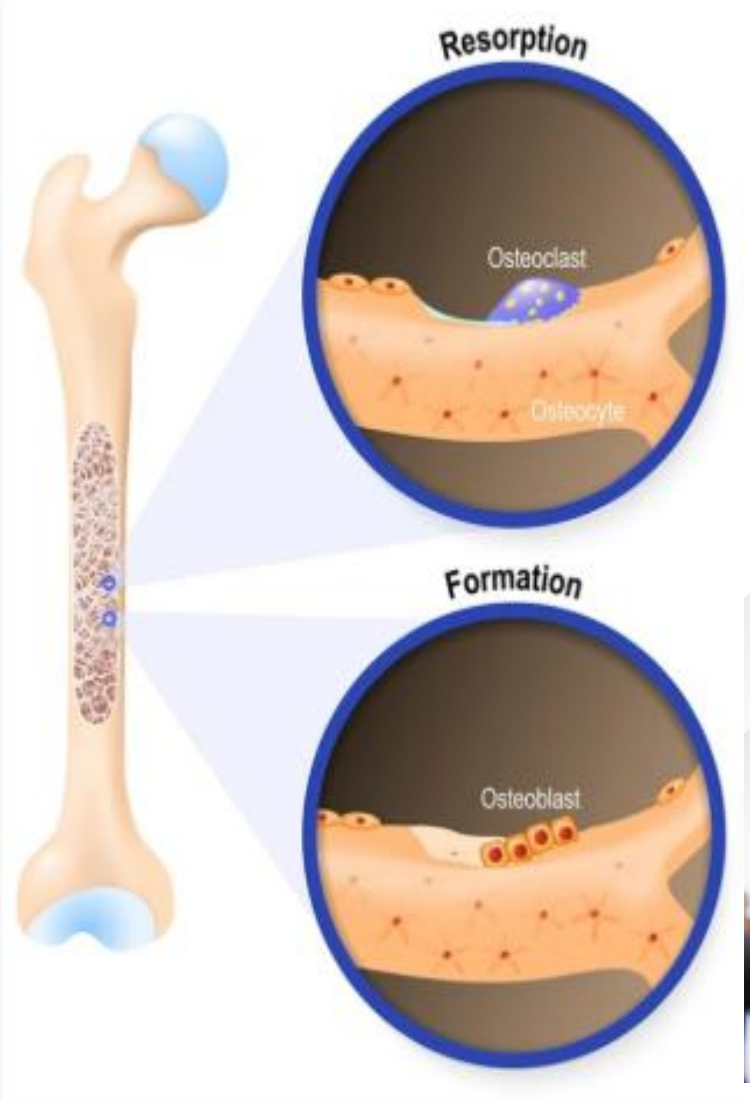


Principle of Bone Scintigraphy

Bones are not inert structures within the human body. Bone continue to change throughout the lifespan.

This process of skeletal change is known as bone **remodeling**, which both protects the structural integrity of the skeletal system and metabolically contributes to the body's balance of calcium and phosphorus.

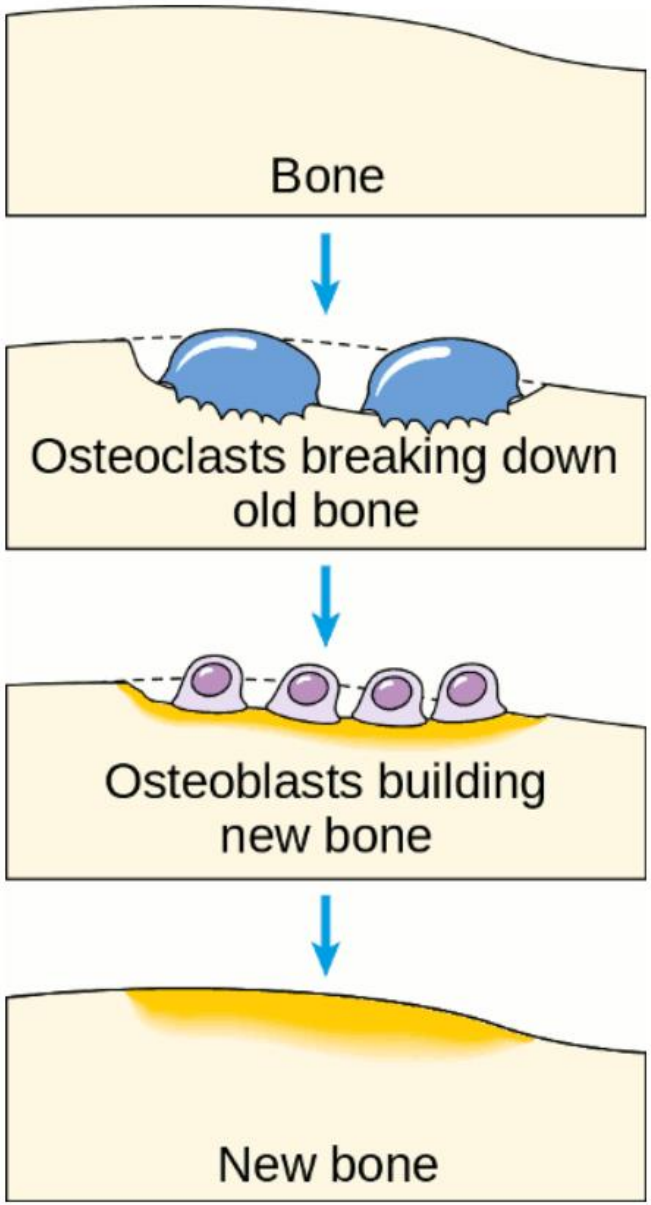
Remodeling entails the resorption of old or damaged bone, followed by the deposition of new bone material.

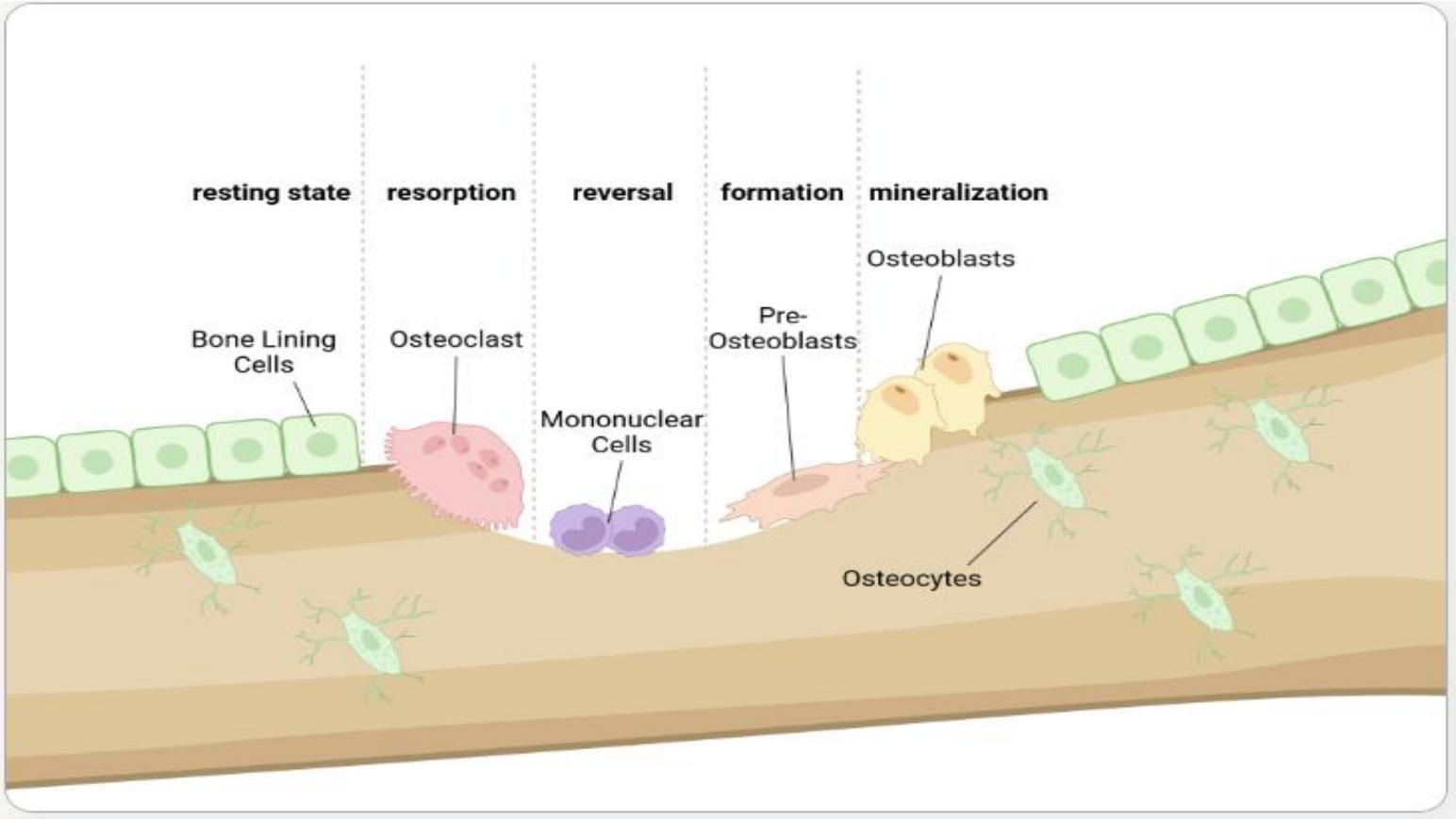


There are 2 primary cells responsible for both the resorption and deposition phases of bone remodeling: osteoclasts and osteoblasts; however, osteocytes also have a role in this process.

The activity of these cells, particularly **the osteoclasts**, is influenced directly or indirectly by hormonal signals.

This interaction between bone remodeling cells and hormones creates the opportunity for a multitude of pathophysiological consequences.





The process of bone remodeling. During the resting state, the bone matrix surface is covered by bone lining cells. When the resorption phase starts, osteoclasts are recruited to resorb bone tissue. Between bone resorption and formation, the reserval phase takes place, in which mononuclear cells migrate to the resorbed site. Osteoblast progenitor cells are recruited to the resorbed bone to differentiate into mature osteoblasts, secreting a mineralized Extracellular matrix (ECM).

Radiopharmaceuticals

The most commonly used radiopharmaceutical is:
99mTc-labeled diphosphonate, particularly:

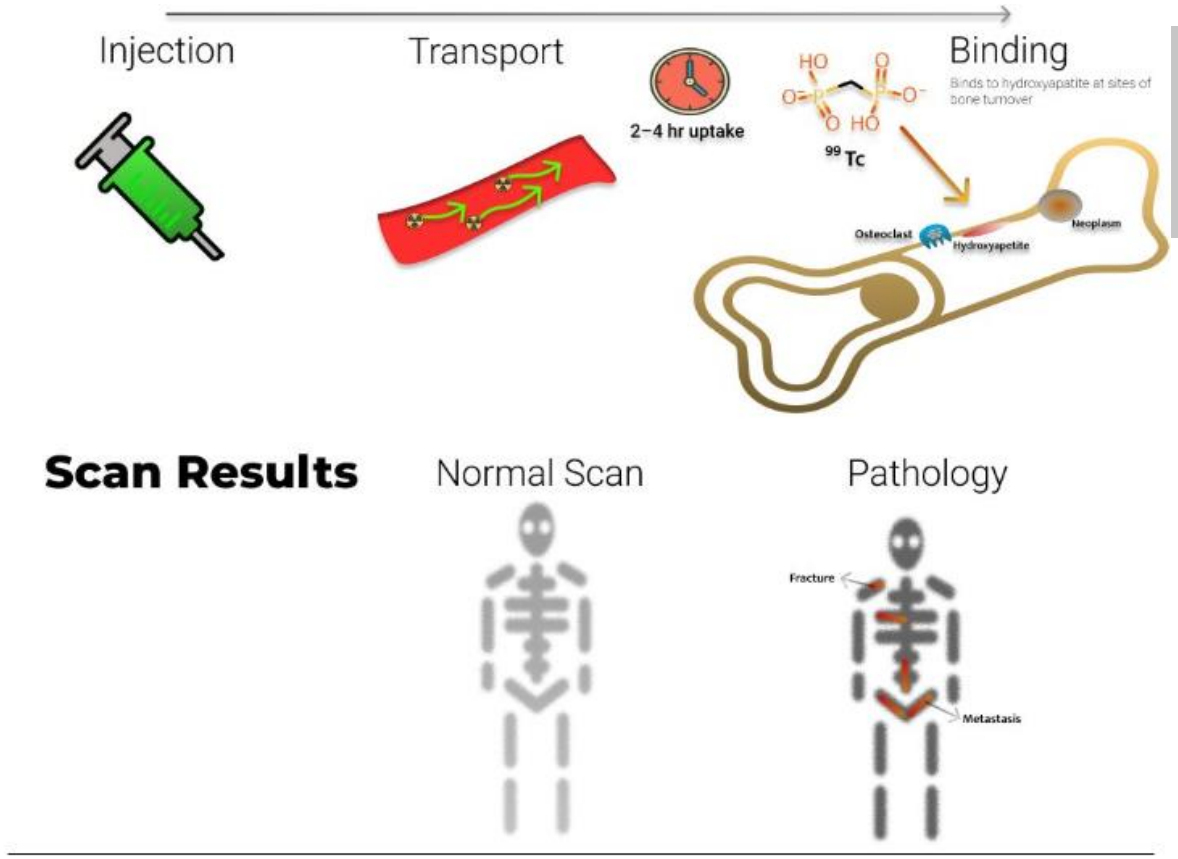
I. 99mTc-MDP – methylene diphosphonate
II. 99mTc-HDP – hydroxymethylene diphosphonate





The radiopharmaceutical accumulates preferentially in areas of increased bone remodeling.





Radiopharmaceutical: 99mTc-MDP
Radionuclide: Technetium-99m
Physical half-life: approximately 6 hours
Principal gamma energy: 140 keV
Imaging: Gamma camera



After intravenous injection:

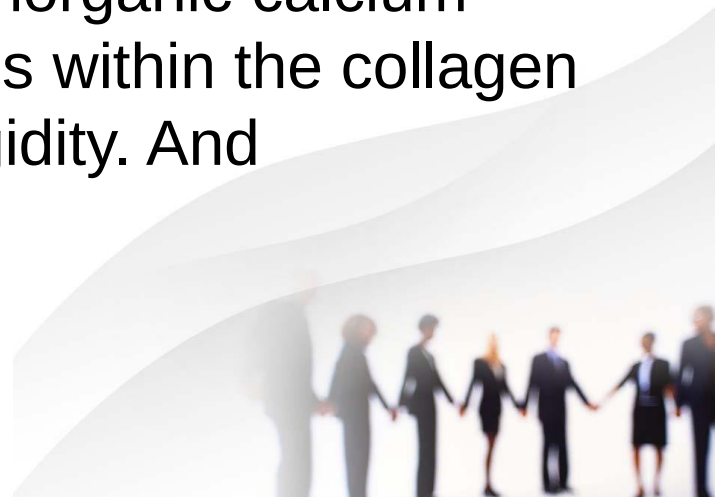
99mTc-diphosphonate → blood circulation → bone → adsorption onto hydroxyapatite crystals

The amount of tracer uptake depends mainly on:

- Blood flow
- Bone blood pool
- Osteoblastic activity
- Surface area of exposed hydroxyapatite

Therefore, a bone scan is primarily a study of bone turnover, rather than simply an anatomical study of bone.

Hydroxyapatite crystals are the natural inorganic calcium phosphate mineral crystal that embededs within the collagen matrix to give bone its hard structure, rigidity. And compressive strength.



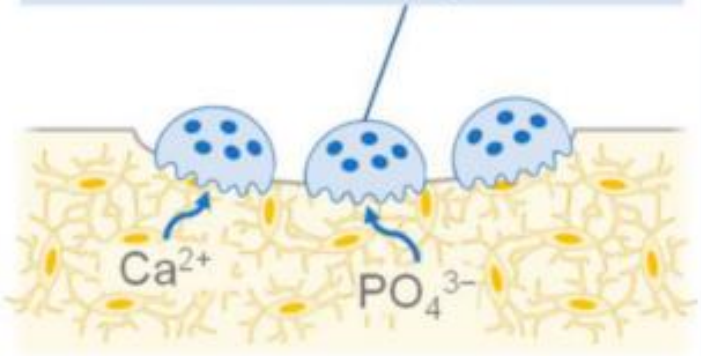
Osteocyte

- Buried osteoblast in the process of bone deposition
- Networks among the cells for detecting mechanical load
- Association with the activity and generation of osteoblasts and osteoclasts



Osteoclast

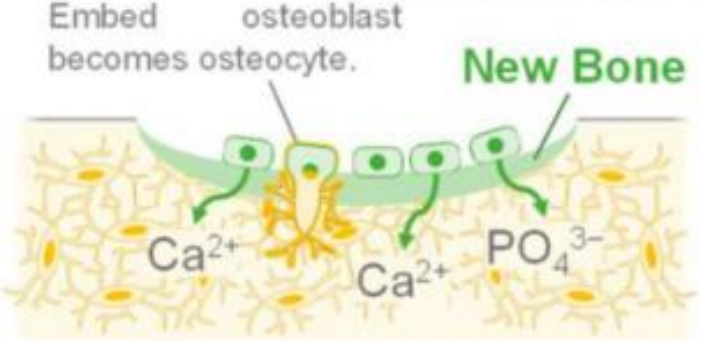
- Phagocytic multinucleated cell
- Function of bone resorption



Activation **Bone resorption**

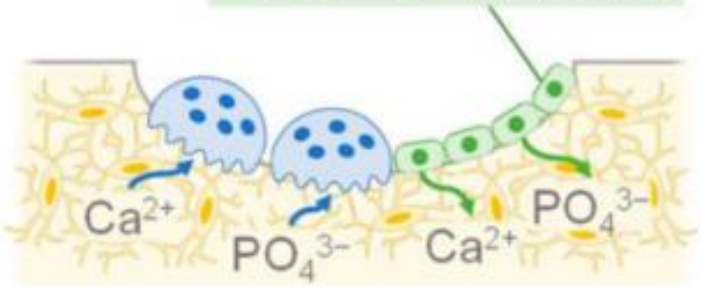
Bone formation **Equilibrium**

Embed osteoblast becomes osteocyte.



Osteoblast

The cell for bone formation.





Scintigraphic imaging with 99mTc-labelled phosphonates detect active bone formation throughout the skeleton.

Active bone formation is altered by many benign and malignant diseases conditions, most frequently by increasing bone formation at the site of infection, in the bone osteoblastic metastasis and new bone formation at sites of trauma.

The bone uptake of these agents reflects a combination of local osteoblastic activity and local arterial blood flow.

Increased uptake = increased osteoblastic activity / bone remodeling. It does not automatically mean malignancy.



Technetium-99 (99mTc)

Metastable Technetium-99 (99mTc) is a well-suited radionuclide for medical imaging. Radiopharmaceuticals based on 99mTc are used worldwide, accounting for about 85% of all nuclear medicine diagnostics. Nuclear medicine departments in hospitals usually use 99mTc as extracted from 99Mo/99mTc generators, where 99Mo, whose half-life is T1/2 = 2.77d , acts as 99mTc precursor. This makes possible the delivery of the generators at long distances from the production sites. At present, 99Mo is almost exclusively obtained from the fission of 235U-containing targets, irradiated in a small number of research nuclear fission reactors in the world.

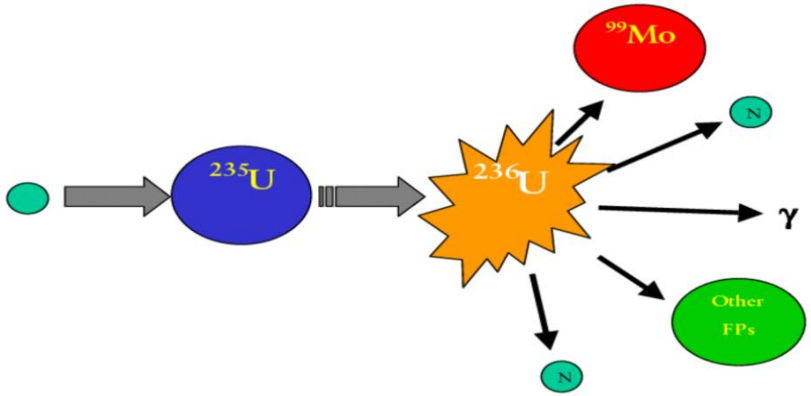
Why 99mTc?

- ✓ Appropriate gamma energy for gamma cameras
- ✓ Short physical half-life
- ✓ Widely available
- ✓ Good bone uptake
- ✓ Relatively low radiation dose

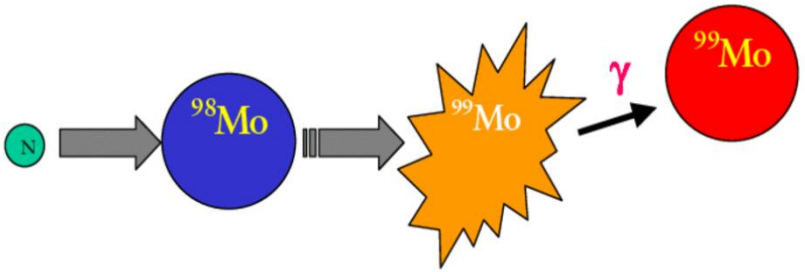


There are two primary processes for producing Molybdenum-99 (Mo99):

I. Fission of uranium-235 (U235).
II. Neutron capture of Molybdenum-98 (Mo98).

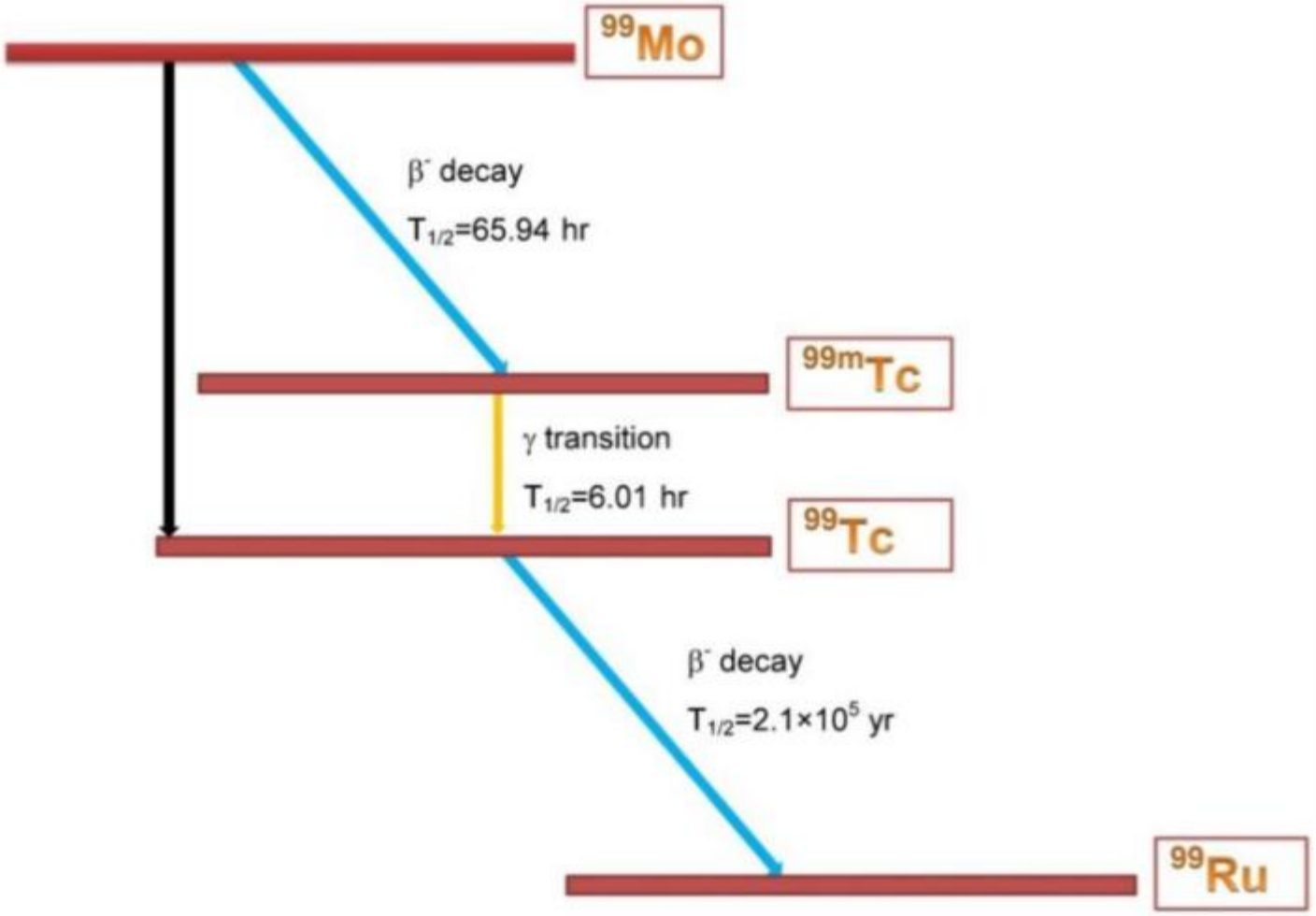


Fission of uranium-235 (U235).



Neutron capture of Molybdenum-98 (Mo98).




^{99}Mo
β⁻ decay
$T_{1/2}$=65.94 hr
^{99m}Tc
γ transition
$T_{1/2}$=6.01 hr
^{99}Tc
β⁻ decay
$T_{1/2}$=2.1×10^5 yr
^{99}Ru

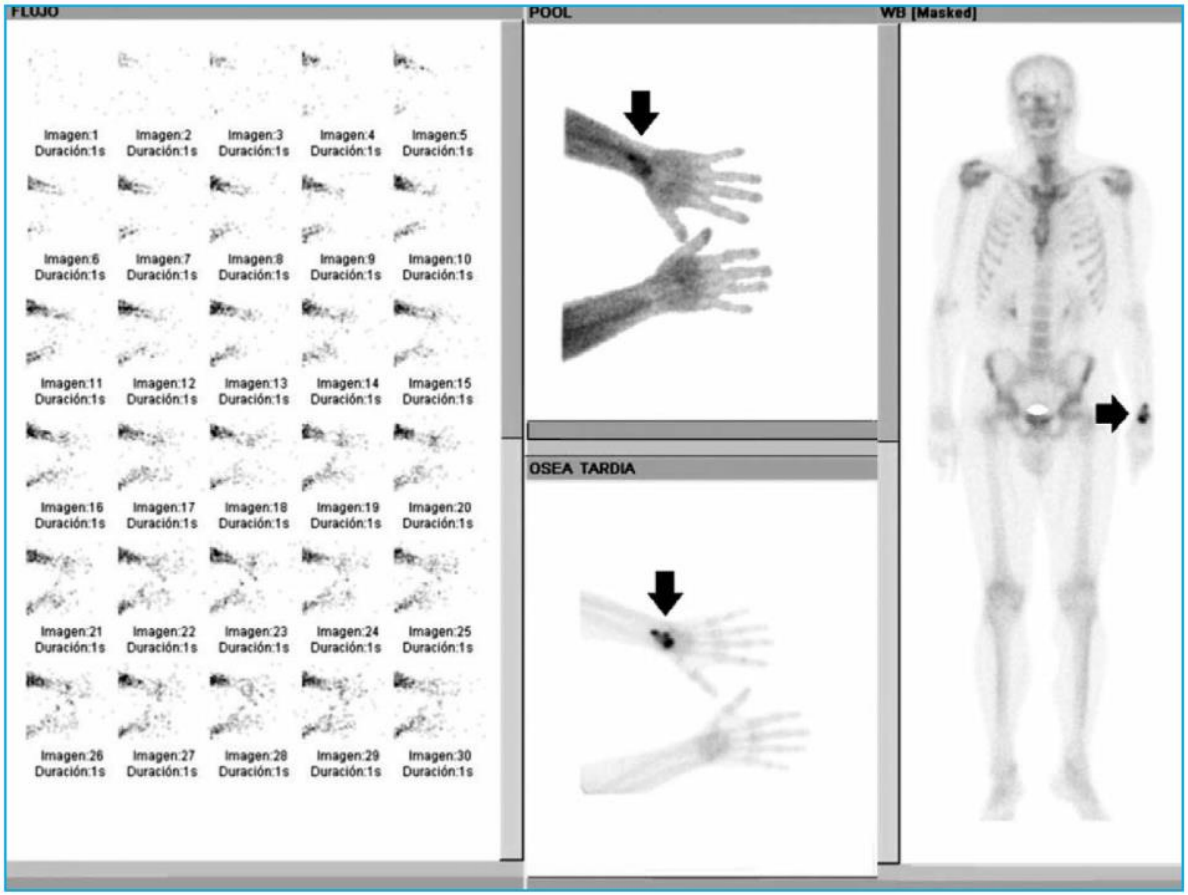


Imaging Protocol

A conventional bone scan can include:

Three-phase bone scan

I. Flow phase
II. Blood-pool phase
III. Delayed skeletal phase



Three-phase bone scintigraphy is designed to separate vascular perfusion, soft-tissue blood-pool activity, and delayed skeletal uptake.

For the technologist, accurate timing and consistent acquisition geometry are critical because interpretation depends on comparing the three phases.





Phase I: Flow / Perfusion Phase

Purpose

The flow phase evaluates arterial blood flow and regional vascularity at the moment of tracer injection.

It is particularly important when evaluating:

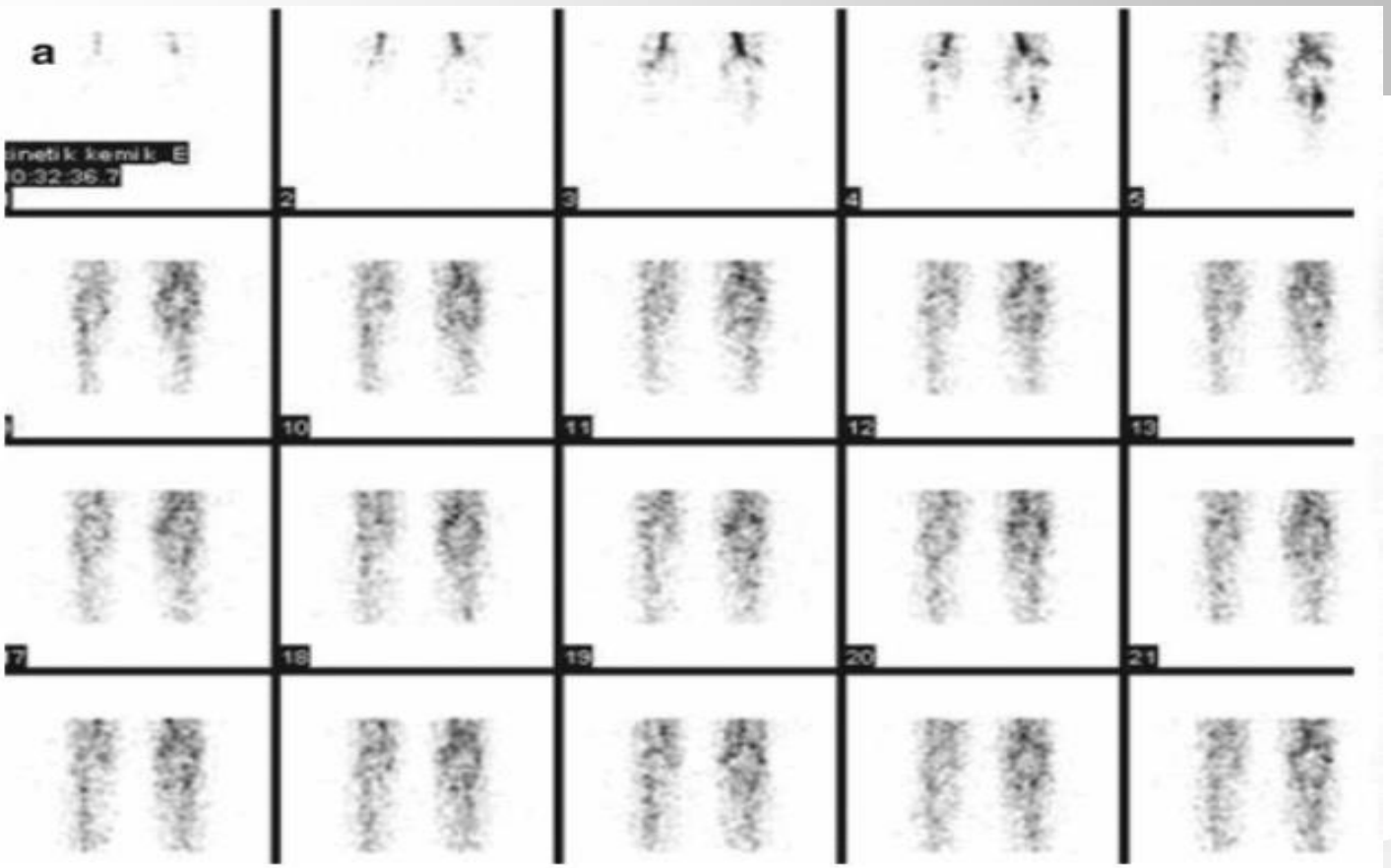
- ✓ Osteomyelitis
- ✓ Cellulitis
- ✓ Acute inflammation
- ✓ Fracture
- ✓ Complex regional pain syndrome
- ✓ Hypervascular tumors
- ✓ Prosthetic-joint complications

Patient and camera positioning

The patient is positioned so that the region of interest is clearly within the gamma-camera field of view.



Phase I: Flow / Perfusion Phase



For example:
Foot/ankle → posterior or plantar/dorsal projection depending on indication
Knee → anterior
Hip → anterior
Hand/wrist → appropriate anterior/posterior or palmar/dorsal view



Phase II: Blood-Pool / Soft-Tissue Phase



The blood-pool phase is obtained immediately after completion of the flow acquisition, generally within several minutes of injection.

What does it represent?

It reflects tracer within:

Blood vessels

Capillary/interstitial space

Hyperemic soft tissues

It therefore provides information about soft-tissue vascularity and hyperemia.

Acquisition

Static planar images

Same projection as the flow phase



Common projections:

- Anterior
- Posterior
- Lateral
- Oblique
- The exact views depend on the anatomical region.

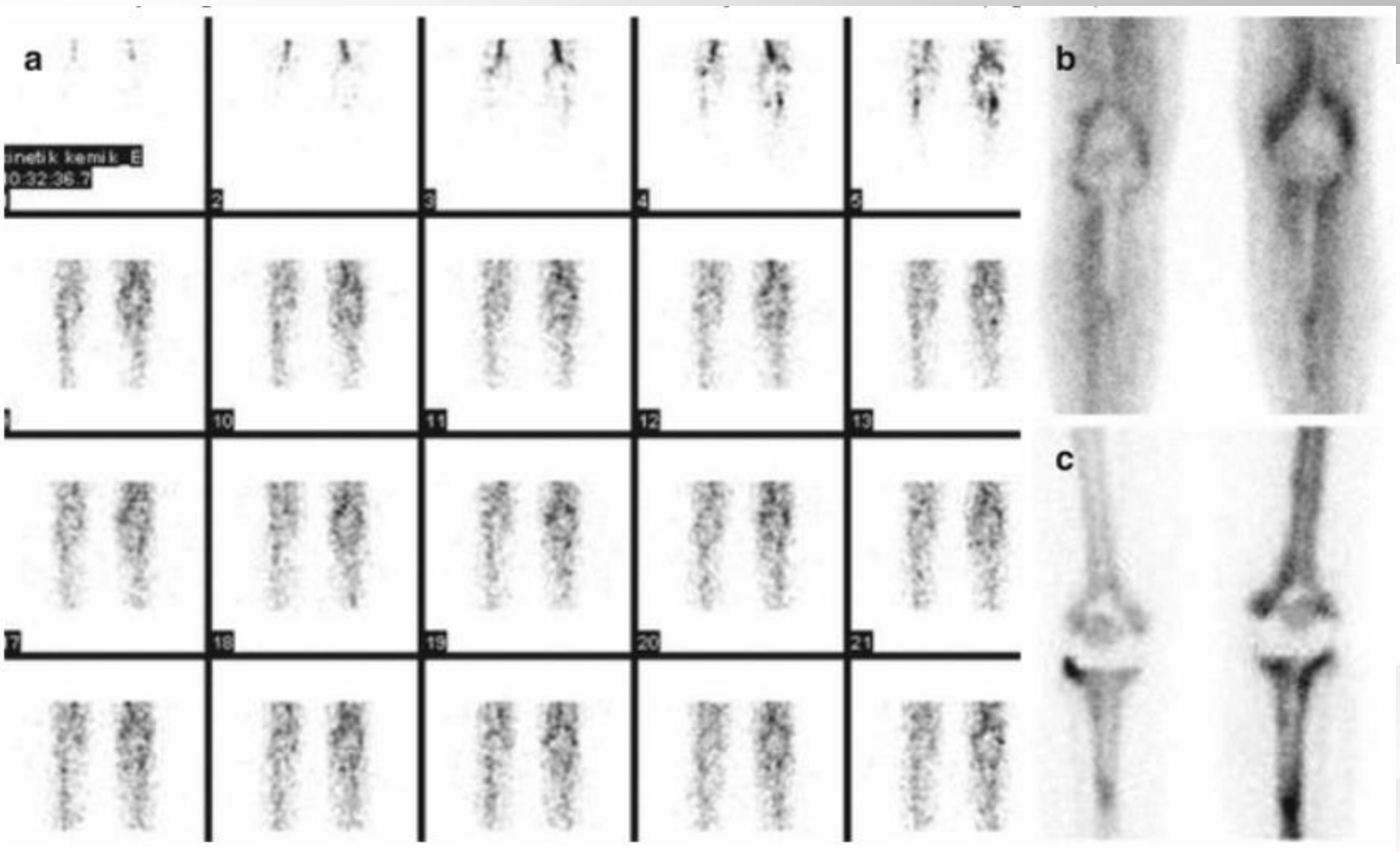
Why use the same positioning?

The flow, blood-pool and delayed images should be spatially comparable.

This makes it easier to determine whether an abnormality is:
vascular → soft tissue → bone
rather than simply appearing in different locations because of patient repositioning.



Flow / Perfusion Phase and Blood-Pool / Soft-Tissue Phase



Phase III: Delayed Skeletal Phase

The delayed phase evaluates bone tracer uptake.

Timing

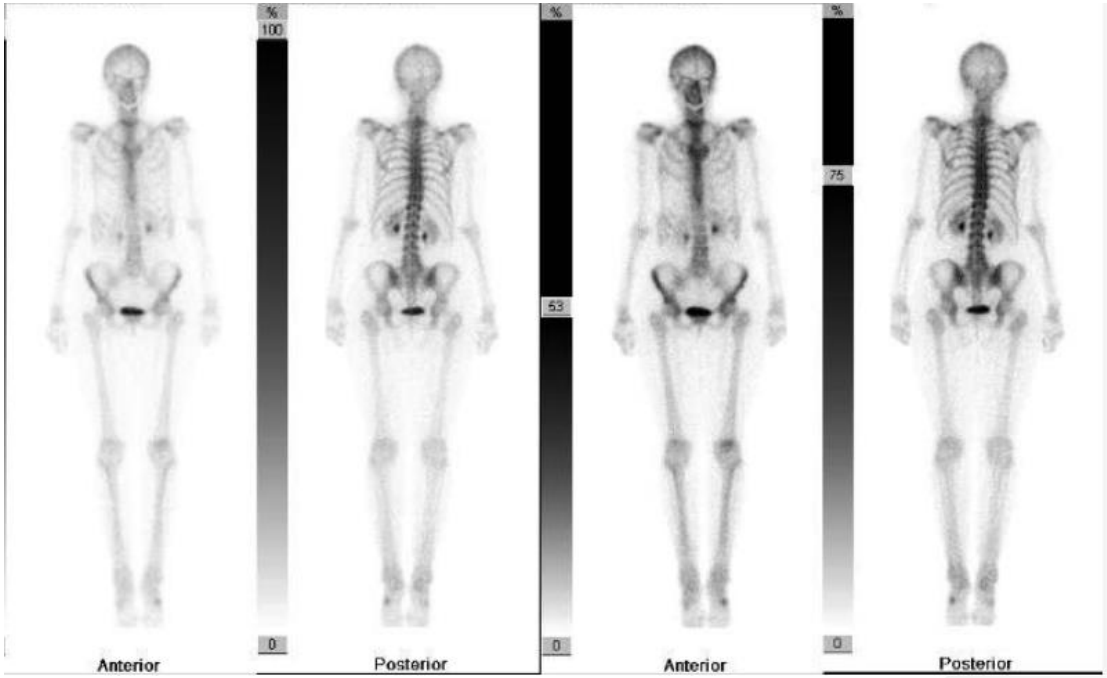
Usually performed approximately:

2–4 hours after injection

The exact timing depends on:

- ❑ Radiopharmaceutical
- ❑ Clinical indication
- ❑ Departmental protocol
- ❑ Patient clearance
- ❑ Need for SPECT/SPECT-CT







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Thank You



